

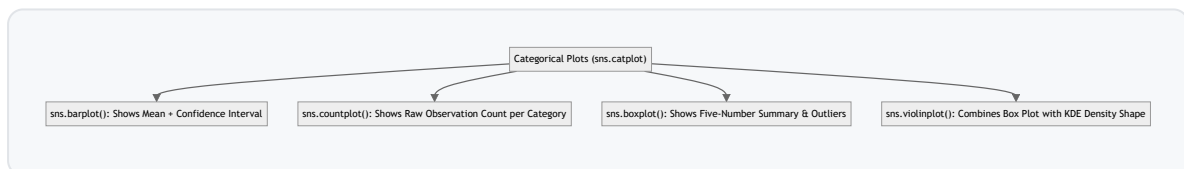
Day 19 - Lecture 19.13: Categorical Plots in Seaborn (`barplot` , `boxplot` , `catplot`)

1. Overview & Purpose

Categorical Plots are used when one variable is **categorical** (discrete groups) and the other is **numerical** (continuous measurement). Common questions answered:

- Which day of the week generates the highest average total bill?
 - Do men and women have different spending distributions across days?
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2. Core Categorical Plot Types



Critical Distinction: `barplot` vs `countplot`

- `sns.barplot(data, x='day', y='total_bill')` : Aggregates numerical values (calculates the **mean** by default) and draws error bars.
 - `sns.countplot(data, x='day')` : Simply counts how many rows exist for each category (frequency count).
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3. Code Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

# 1. Bar Plot: Average Total Bill by Day and Sex
plt.figure(figsize=(8, 5))
sns.barplot(data=tips, x="day", y="total_bill", hue="sex", palette="Blues", error=
plt.title("Average Total Bill by Day and Sex (barplot)", fontsize=13, fontweight=
plt.xlabel("Day of Week")
plt.ylabel("Mean Total Bill ($)")
plt.show()

# 2. Box Plot: Distribution of Total Bill by Day
plt.figure(figsize=(8, 5))
sns.boxplot(data=tips, x="day", y="total_bill", palette="Set3")
plt.title("Spread & Outliers of Total Bill Across Days (boxplot)", fontsize=13, f
plt.xlabel("Day of Week")
plt.ylabel("Total Bill ($)")
plt.show()

# 3. Grouped Box Plot with hue
plt.figure(figsize=(9, 5))
sns.boxplot(data=tips, x="day", y="total_bill", hue="sex", palette="Pastel1")
plt.title("Box Plot Grouped by Day and Sex", fontsize=13, fontweight="bold")
plt.legend(title="Sex", loc="upper left")
plt.show()
```

4. Key Takeaways

- Seaborn's `barplot` calculates statistical aggregates (**mean**) automatically—do not confuse it with a Matplotlib bar chart which plots raw values passed to it.
- Adding `hue="sex"` automatically groups and dodges the bars or boxes side by side.